Ultrasound Imaging

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Content current as of:
09/19/2024

Description

Ultrasound imaging (sonography) uses high-frequency sound waves to view inside the body. Because ultrasound images are captured in real-time, they can also show movement of the body's internal organs as well as blood flowing through the blood vessels. Unlike X-ray imaging, there is no ionizing radiation exposure associated with ultrasound imaging.

In an ultrasound exam, a transducer (probe) is placed directly on the skin or inside a body opening. A thin layer of gel is applied to the skin so that the ultrasound waves are transmitted from the transducer through the gel into the body.

Picture of a transducer (probe) used during an ultrasound exam.

The ultrasound image is produced based on the reflection of the waves off of the body structures. The strength (amplitude) of the sound signal and the time it takes for the wave to travel through the body provide the information necessary to produce an image.

Uses

Ultrasound imaging is a medical tool that can help a physician evaluate, diagnose and treat medical conditions. Common ultrasound imaging procedures include:

- Abdominal ultrasound (to visualize abdominal tissues and organs)
- Bone sonometry (to assess bone fragility)
- Breast ultrasound (to visualize breast tissue)
- Doppler fetal heart rate monitors (to listen to the fetal heart beat)
- Doppler ultrasound (to visualize blood flow through a blood vessel, organs, or other structures)
- Echocardiogram (to view the heart)
- Fetal ultrasound (to view the fetus in pregnancy)
- Ultrasound-guided biopsies (to collect a sample of tissue)
- Ophthalmic ultrasound (to visualize ocular structures
- Ultrasound-guided needle placement (in blood vessels or other tissues of interest)

Benefits/Risks

Ultrasound imaging has been used for over 20 years and has an excellent safety record. It is based on non-ionizing radiation, so it does not have the same risks as X-rays or other types of imaging systems that use ionizing radiation.

Although ultrasound imaging is generally considered safe when used prudently by appropriately trained health care providers, ultrasound energy has the potential to produce biological effects on the body. Ultrasound waves can heat the tissues slightly. In some cases, it can also produce small pockets of gas in body fluids or tissues (cavitation). The long-term consequences of these effects are still unknown. Because of the particular concern for effects on the fetus, organizations such as the American Institute of Ultrasound in Medicine have advocated prudent use of ultrasound imaging in pregnancy. Furthermore, the use of ultrasound solely for non-medical purposes such as obtaining fetal 'keepsake' videos has been discouraged. Keepsake images or videos are reasonable if they are produced during a medically-indicated exam, and if no additional exposure is required.

Information for Patients including Expectant Mothers

For all medical imaging procedures, the FDA recommends that patients talk to their health care provider to understand the reason for the examination, the medical information that will be obtained, the potential risks, and how the results will be used to manage the medical condition or pregnancy. Because ultrasound is not based on ionizing radiation, it is particularly useful for women of child-bearing age when CT or other imaging methods would otherwise result in exposure to radiation.

Expectant Mothers

Ultrasound is the most widely used medical imaging method for viewing the fetus during pregnancy. Routine examinations are performed to assess and monitor the health status of the fetus and mother. Ultrasound examinations provide parents with a valuable opportunity to view and hear the heartbeat of the fetus, bond with the unborn baby, and capture images to share with family and friends.

In fetal ultrasound, three-dimensional (3D) ultrasound allows the visualization of some facial features and possibly other parts such as fingers and toes of the fetus. Four-dimensional (4D) ultrasound is 3D ultrasound in motion. While ultrasound is generally considered to be safe with very low risks, the risks may increase with unnecessary prolonged exposure to ultrasound energy, or when untrained users operate the device.

Expectant mothers should also be aware of concerns with purchasing over-the-counter fetal heartbeat monitoring systems (also called doptones). These devices should only be used by trained health care providers when medically necessary. Use of these devices by untrained persons could expose the fetus to prolonged and unsafe energy levels, or could provide information that is interpreted incorrectly by the user.

Additional resources on ultrasound imaging:

- RadiologyInfo.org: Obstetrical Ultrasound
- Image Gently: The Alliance for Radiation Safety in Pediatric Imaging

Information for Health Care Providers

As part of FDA's Initiative To Reduce Unnecessary Radiation Exposure From Medical Imaging, FDA has recommended that health care providers consider examinations that use little or no ionizing radiation exposure, such as ultrasound or MRI, if medically appropriate. Ultrasound imaging does introduce energy into the body, and laboratory studies have shown that diagnostic levels of ultrasound can produce physical effects in tissue, such as pressure oscillations with subsequent mechanical effects and rise in temperature. Therefore, FDA recommends that health care providers consider ways to minimize exposure while maintaining diagnostic quality when using ultrasound. As with all other imaging modalities, the principles of As Low As Reasonably Achievable (ALARA) should be practiced by health care providers.

Individual states regulate the use of diagnostic ultrasound through recommendations and requirements for personnel qualifications, quality assurance and quality control programs and facility accreditation.

Information for Facilities Conducting Ultrasound Imaging

Ultrasound practices should consider site and staff participation in voluntary accreditation and certification programs that address both safety and effectiveness of the device following the principles of As Low As Reasonably Achievable (ALARA), such as those offered by the American Institute of Ultrasound in Medicine and the American Registry of Diagnostic Medical Sonographers.

Any health care facility employing ultrasound should conduct regular quality control tests to ensure that equipment is functioning properly.

Information for Industry: Ultrasound Manufacturers

Manufacturers of electronic radiation emitting products sold in the United States are responsible for compliance with the Federal Food, Drug and Cosmetic Act (FFDCA), Chapter V, Subchapter C - Electronic Product Radiation Control.

Manufacturers of ultrasound imaging products are responsible for compliance with all applicable requirements of Title 21 Code of Federal Regulations (Subchapter J, Radiological Health) Parts 1000 through 1005:

1000 - General

1002 - Records and Reports

1003 - Notification of defects or failure to comply

1004 - Repurchase, repairs, or replacement of electronic products

1005 - Importation of electronic products

There are no federal radiation safety performance standards for diagnostic ultrasound.

Because they are medical devices, ultrasound imaging equipment must also comply with the medical device regulations. For more information, see Getting to Market with a Medical Device.

Industry Guidance - Documents of Interest

- Guidance for Industry and FDA Staff - Information for Manufacturers Seeking Marketing Clearance of Diagnostic Ultrasound Systems and Transducers (PDF Version) (PDF - 474KB)
- Declaration for Imported Electronic Products Subject to Radiation Control Standards (PDF - 399KB)
- Ultrasound Letter to Industry
- Variance Application Process

Reporting Problems to FDA

Prompt reporting of adverse events can help the FDA identify and better understand the risks associated with the product. We encourage health care providers and patients who suspect a problem with a medical imaging device to file a voluntary report through MedWatch, the FDA Safety Information and Adverse Event Reporting Program.

Health care personnel employed by facilities that are subject to FDA's user facility reporting requirements should follow the reporting procedures established by their facilities.

Medical device manufacturers, distributors, importers, and device user facilities (which include many health care facilities) must comply with the Medical Device Reporting (MDR) Regulations of 21 CFR Part 803.

In addition to following the general recommendations (for manufacturers, facilities, and any member of the public) for reporting problems for adverse events associated with ultrasound imaging, the following information should be included in reports, if available:

- the protocol being followed during the event;
- sample images, if available;
- the conditions of operation, including technical parameters such as:
  - mode of operation
  - clinical application
  - Thermal Index (TI) and Mechanical Index (MI)
  - duration of scan, if known

Other Resources

- Does the Product Emit Radiation?
- Getting a Radiation Emitting Product to Market
- Records and Reporting (Radiation-Emitting Products)
- Importing and Exporting Electronic Products

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